

Negative Exponents with Variable BasesIndependent Set 3.T2 **ANSWER KEY***Six questions, on your own, in silence. Show the step that does the work.*

1. Rewrite using only positive exponents.
- m^{-6}

$$\frac{1}{m^6}$$

2. Rewrite using only positive exponents.
- $5y^{-3}$

$$\frac{5}{y^3}$$

The 5 stays upstairs.

3. Rewrite using only positive exponents.
- $\frac{1}{x^{-7}}$

$$x^7$$

The factor moves up.

4. Simplify. Use positive exponents only.
- $x^2 \cdot x^{-9}$

$$\frac{1}{x^7}$$

$$2 + (-9) = -7.$$

5. Simplify. Use positive exponents only.
- $(3m^{-2})^3$

$$\frac{27}{m^6}$$

The 3 is cubed as well.

6. Find the value of
- n
- .
- $y^{-3} \cdot y^n = y^4$

$$n = 7$$

$$-3 + n = 4.$$